VPN and Tunneling

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ITIS 6167: Network Security

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March 21, 2024

Introduction

The purpose of this lab is to study the effectiveness of Virtual Private Network (VPN) and how computers can communicate securely on them. By employing the use of Wireshark, we will be able to perform an analysis on the network packets and this information is encapsulated. Many services inside private networks utilize IP addresses as a means for authorization. However, using IP addresses is not secure anymore, because it would mean opening too many access points or "doors", providing attackers with opportunities to access the protected resources (Du 2022, p.170). That being the case many organizations utilize an encryption layer in their VPN, this lab however will not contain this feature. We will be using VirtualBox, an open-source software virtualizing tool, to run the host operating system (OS) located in the provided file "SEED Ubuntu16.04 VM (32-bit)". We will then construct and configure two VPNs, one to be the client and the other to be the VPN server to demonstrate the implementation of the IP Tunnel. Next, we will use Wireshark, an open-source network analyzer tool, to capture the network traffic of the VPN, revealing the attack vectors that threat actors could possibly maneuver. Finally, we will use Wireshark to analyze VPN IP Tunneling and its effect on network traffic (Univsersity of North Carolina at Charlotte , 2024).

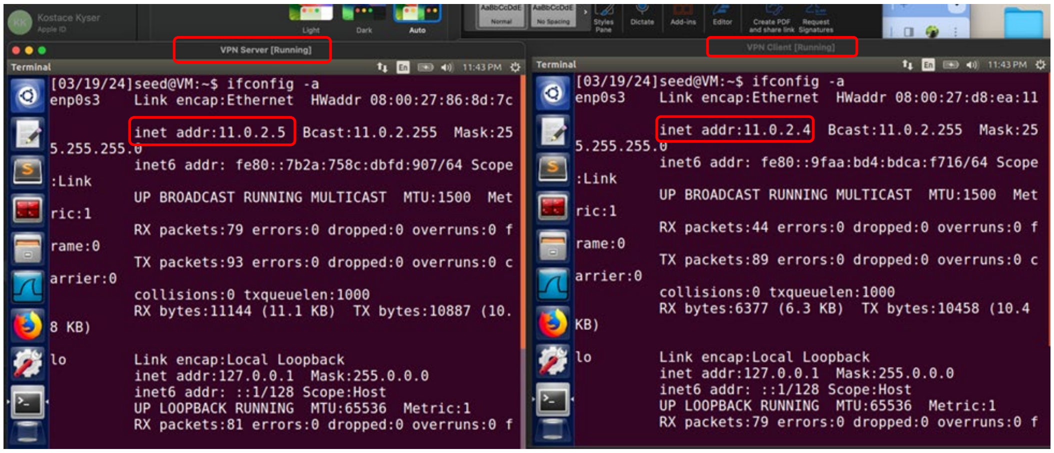
VPN Tunnel

"VPN is a private and secure connection [1] between two or more networks or computers who share protected data, using a single secure channel between the endpoints, over a public data network (for example WAN) or through the Internet. Tunneling represents the ability to make circuit oriented connections in WAN topologies oriented on packets. This process is the main technical concept of VPN" (Daniel Simion, 2013). The primary goal of a VPN is transparency, the communication with the host should always be recure. To achieve this a VPN must be implemented on the network layer (layer 3) or the data link layer (layer 2). A VPN residing at the network layer is called a layer 3 VPN. "Known to be the strongest authentication and encryption method IPSec works at layer 3 of the OSI network model" (Daniel Simion, 2013 p. 2). Since "encrypted packets will not be able to travel through the Internet to reach their destinations, because routers cannot read the header of the packets", IP Tunneling is used to feed packets into one part of the tunnel and it appears at the other end, keeping them secure in the process. Both ends of the tunnel use the Internet Protocol Security (IPSec) Tunneling and TLS/SSL Tunneling solutions which reside on the IP layer on top TCP/UDP. (Du, 2022 p. 172). Implementing IP tunneling we construct a dedicated virtual line between two computers, a VPN client, and a VPN server. The primary roles of the VPN client and server are to establish a secure tunnel, forward packets through the tunnel and receive said packets on the other end of the tunnel, so packets can securely reach their destinations. (Du, 2022)

We begin the experiment by starting both VPN Server and VPN Client machines. Then we execute the command "ifconfig -a" in the terminal to examine the VPN Client's IP address as 11.0.2.4 and the VPN Server's IP address as 11.0.2.5. being equipped with respective IP addresses we enter the command "ping" with the hostname, to demonstrate the established virtual network interfaces functionality as shown in figure 1. "Since the VPN client's eth1 is in the promiscuous mode, it should be able to see the ping packet" (Du, 2022 p. 195)

Figure 1

Screenshot of VPN Client and Server enp0s3 successfully pinging each other



Note. This was found using the "ifconfig -a" command in terminal.

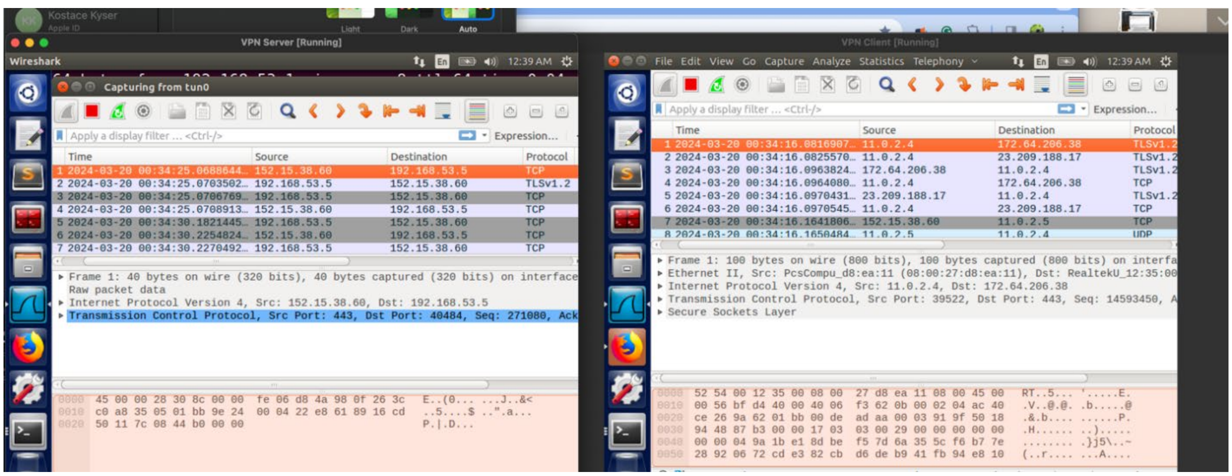
A Virtual Network Interface is a virtual representation of a computer's network interface, one example of this type of interface is the TUN/TAP interface. These interfaces can be seen as a point-to-point network device, connecting the two computers (Du, 2022). "Point-to-Point Tunneling Protocol (PPTP) encapsulates PPP frames for transmission over IP internet works in IP datagrams. For tunnel maintenance a TCP connection is used in PPTP, in order to encapsulate PPP frames in tunnelled data is used a modified version of GRE (Generic Routing Encapsulation). The content encapsulated with PPP frames can be compressed or encrypted" (Daniel Simion, 2013 p.2). The TUN device works on the IP layer and supports point-to-point (P2P) network communication. "Linux kernels come with pre-build support for TUN/TAP devices" (Du,2022 p. 178). The TUN/TAP device is used to construct our Layer 3 VPN, configure it, and implement network communication between the VPN Client and VPN Server.

Wireshark is able to capture and analyze IP addresses from the local host, for that reason we were able to obtain several unique IP addresses on the network which are different from the VPN Client and VPN Server's IP address. This is due to the routing configuration, as we observe figure 2, here the ICMP packet is routed to the TUN interface, changing the source IP to the one that is assigned to tun0.

In turn the tunnel application receives the receives the ICMP packet, feeding it to the tunnel, putting it inside a UDP packet toward the VPN Server. As it goes out from the client's normal interface, its source IP address is 11.0.2.4, which is the client's ethernet address. As the UDP packet is returned from the VPN Server, inside we observe an encapsulated ICMP echo reply packet from 192.168.53.5. The tunnel application on the VPN client gets this UDP packet, takes off the encapsulated ICMP packet, and gives the packet to the kernel utilizing tun0 interface.

Figure 2

Screenshot of Wireshark on VPN Server and VPN Client



Note: This is demonstrating TUN/TAP device in IP tunneling

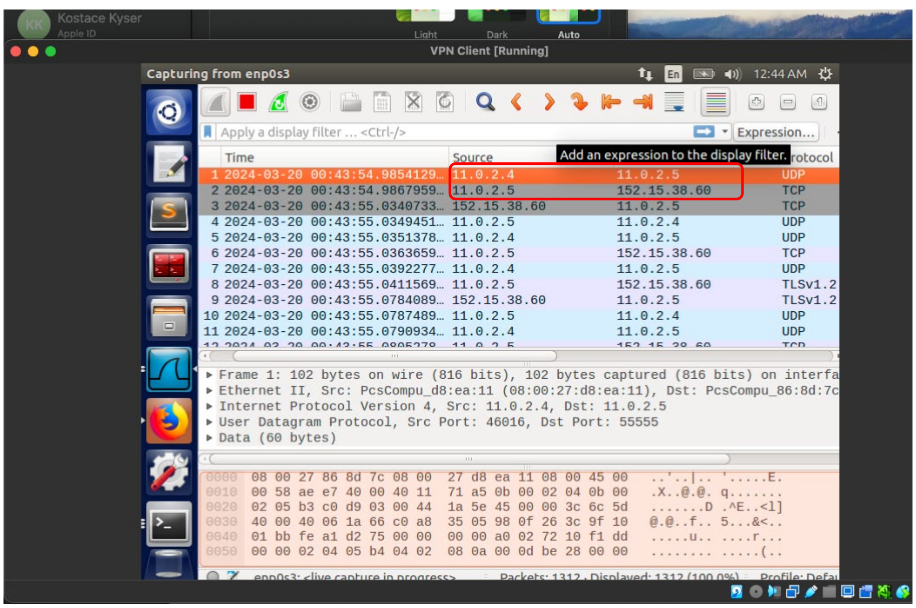
IP Tunneling and Firewalls

Firewalls are important security technologies that are widely deployed in networks, restricting networks traffic. There are situations that may occur where firewalls are too restrictive. "There are many ways to evade firewalls. A typical approach is tunnelling technique, which hides the real purpose of network traffic" (Du, 2022 p. 202). By using VPN tunnelling we are able to bypass the ingress filtering capabilities of firewalls. Firewalls inspect the source and/ or destinations address of each packet, dropping unauthorized invalid packets. VPNs mitigate this feature by hiding the packets contents in the tunnel. Since the tunnel is encrypted, the firewalls are unable to read the contents and thus unable to appropriately filter.

To bypass the firewalls ingress, we use the VPN and form a tunnel between the VPN Client and the VPN Server, a machine outside of the firewall's rules. As shown in figures 3 and 4, from Wireshark, we are able to observe that our VPN Server has forwarded the packets, with a reply. Demonstrating the ability to successfully bypass, all packets going through this interface will have their source IP address changed to the TUN interface's IP address, as shown in figure 4. (Du, 2022)

Figure 3

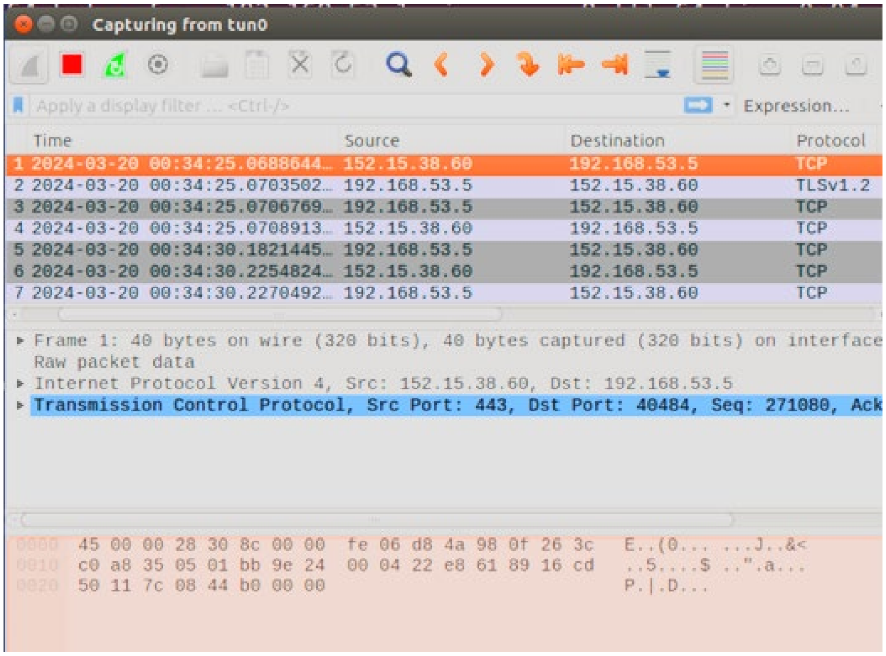
Screenshot of Wireshark capturing enp0s3 interface



Note: VPN Client accessing block side via VPN

Figure 4

Screenshot of Wireshark capturing TUN interface



Note: Client IP address is hidden and replaced with TUN IP address

References

Bullock, J. a. (2017). *Wireshark for Professionals: Using Wireshark and the Metasploits Framework.* Retrieved from Atkins Library : http://ebookcentral.proquest.com/lib/uncc-ebooks/detail.action?docID=4816492

Daniel Simion, M. F. (2013). *Efficiency Consideration for Data Packets Encryption within Wireless VPN Tunneling for Video Streaming.* Suceava: INT J COMPUT COMMUN, ISSN 1841-9836.

Du, W. (2022). *Internet Security A Hands-on Approach.* Syracuse: Independently published .

Julien Bringer, H. C. (2008, 8 26). *Trusted-HB: A Low-Cost Version of HB + Secure Against Man-in-the-Middle Attacks.* Retrieved from https://ieeexplore.ieee.org/document/4608956: https://library.charlotte.edu/

Mahmmod, K. F., Azeez, M. M., & Ahmed, M. A. (2021). IPsec Cryptography for Data Packets Security within VPN Tunneling Networks Communications. *International Conference on Electrical Engineering and Informatics (ICELTICs)* (p. 13). Aceh: IEEEXplore.

Univsersity of North Carolina at Charlotte . (2024, 2 11). *Lab 2 - VPN Project*. Retrieved from UNCC Canvas: https://uncc.instructure.com/courses/217874/assignments/2048974

Walley, W. (2001, Jul/Aug 26). *VPN advantages*. Retrieved from ProQuest: https://www.proquest.com/trade-journals/vpn-advantages/docview/227936273/se-2?accountid=14605